

ADVANCED WEB PROGRAMMING LAB
(Common to CSE-AI&ML, CSE-CS and CSE-DS)

Instruction	: 3 Periods/week	Continuous Internal Evaluation	: 40 Marks
Tutorial	: -	Semester End Examination	: 60 Marks
Credits	: 1.5	Semester End Exam Duration	: 3 Hours

Course Objectives:

- 1 : Able to develop web applications and learn the principles of server-side scripting with Node.js
- 2 : Understanding Components and Event handling in React, create react applications and React forms
- 3 : Working with applications consuming REST APIs

Exercises:

1. Create a git repository and clone it for changes and publish the changes using git bash.
2. Create a real time database in firebase for the student management system and explore the features of Firebase Real Time Database.
3. Explore the various node modules: os, http, etc.
4. Create a REST API and perform various CRUD operations on that.
5. Develop an express web application that can interact with a service to perform CRUD operations on student data. (Use Postman)
6. For the above application create authorized end points using JWT (JSON Web Token).
7. Create a react application using create-react-app and display "Hello World". (ReactJS environment Setup using Node Package Manager, Motivation for using React, Virtual DOM vs Real DOM).
8. Create a react application to render the element with the current system date and time. (Introduction to JSX and its uses, Expressions in JSX, Rendering Elements in React.JS. React DOM. render() function. How does render function work and update DOM)
9. a. Create a functional component that accept the arbitrary inputs (called props) and display the value.
b. Create a class component that accept the arbitrary inputs (called props) and display the value. (Introduction to Components in React, Types of Components in react, Functional vs Class Component, Passing the props to child Components).
10. Create a class component that accept the date in the form of State and displays the date. (Passing the state to Class Component, Props vs State).
11. Create a react application to demonstrate the component lifecycle. (Understanding the Lifecycle Methods: Phases of React Component Lifecycle, React Component Lifecycle methods)
12. Create a react application to create a two buttons and handle the click events. The click event of first display the current system date and time using event listener (Normal functions). For click event of second button display "Good Bye" by using event listener (arrow functions) (Handling the events in React, Binding event listeners)
13. Create a react application to implement the calculator application by creating UI and providing the functionality using events. Also make it look good by using styles. (React Styling: How to use the Styles in React)
14. Create a react application to create a registration page using forms as controlled components. (Working with the React Forms)
15. Create a react application to understand how the React Router works. The application will contain four components: home component, product component, about component, and contact component. We will use React Router to navigate between these components. (React Routing, React Router components)

16. Create a simple e-commerce storefront for a bookstore which consists of four screens:
 - Home —The storefront with a book list
 - Product page —A separate product page
 - Cart —A web page showing the quantities and titles selected by the user.
 - Checkout—A print-ready invoice with the list of books
17. Access a web service in react that fetches the weather information from open weather map and the display the current and historical weather information using graphical representation using chart.js.
 - (Consuming the REST APIs in the React)
18. Create a TODO application in react with necessary components and deploy it into github.

Course Outcomes: At the end of the course, the student should be able to

- CO 1 : To design and build robust REST APIs using Node.js, demonstrate the Express Framework
- CO 2 : Understand components in React
- CO 3 : Handle event in React and bind event listeners
- CO 4 : Work with React forms
- CO 5 : Create applications for consuming REST APIs

References:

1. Azat Mardan, React Quickly, Manning Publications, August 2017.
2. , Full Stack Web App Development with Mongo, Express, React and Node, Vasan Subramanian, Pro MERN Stack, 2nd Edition, APress, 2019.